

KENDRIYA VIDYALAYA SANGTHAN VARANASI REGION

PRE-BOARD-I EXAMINATION (2024 –25)

CLASS- XII

Maximum marks- 70

SUBJECT- PHYSICS

Maximum time- 3 Hours

General Instructions:

- 1) There are 33 questions in all. All questions are compulsory.
- 2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- 3) All the sections are compulsory.
- 4) Section A contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each,
Section B contains five questions of two marks each,
Section C contains seven questions of three marks each,
Section D contains two case study-based questions of four marks each and
Section E contains three long answer questions of five marks each.
- 5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E.
- 6) Use of calculators is not allowed.
- 7) You may use the following values of physical constants where ever necessary:

$$c = 3 \times 10^8 \text{ m/s}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$$

$$M_p = 1.673 \times 10^{-27} \text{ kg}$$

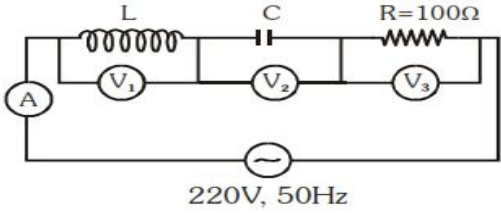
$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$$

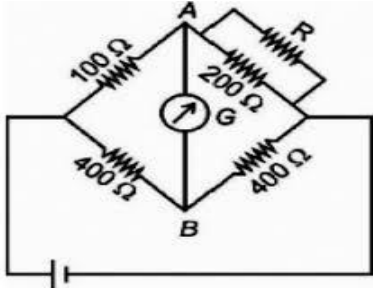
$$M_e = 9.1 \times 10^{-31} \text{ kg}$$

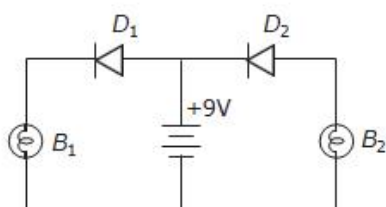
$$M_n = 1.675 \times 10^{-27} \text{ kg}$$

Q.No	Questions	Marks
Section A		
1.	Force between charges A and B is F. If 75% charge of A is transferred to B then force between A and B is :  (a) $\frac{F}{4}$ (b) F (c) 4F (d) None of these	1
2.	On applying an electric field of $5 \times 10^{-8} \text{ Vm}^{-1}$ across a conductor, the current density through it is 2.5 Am^{-2} The resistivity of the conductor is: (a) $1 \times 10^{-8} \Omega \text{ m}$ (b) $2 \times 10^{-8} \Omega \text{ m}$ (c) $0.5 \times 10^{-8} \Omega \text{ m}$ (d) $12.5 \times 10^{-8} \Omega \text{ m}$	1
3.	Which of the following is not the property of an equipotential surface? (a) They do not cross each other. (b) The work done in carrying a charge from one point to another on an equipotential surface is zero. (c) For a uniform electric field, they are concentric spheres. (d) They can be imaginary spheres.	1

4.	Beams of electrons and protons move parallel to each other in the same direction, They: (a) Attract each other (b) Repel each other (c) Neither attract nor repel (d) Force of attraction or repulsion depends upon speed of beams.	1
5.	In the given circuit the reading of voltmeter V_1 and V_2 are 300 volts each. The reading of the voltmeter V_3 and ammeter A are respectively:  (a) 220 V, 2.2 A (b) 150 V, 2.2 A (c) 100 V, 2.0 A (d) 220 V, 2.0 A	1
6.	Amplitude of electric field in EM wave is give as $E_0 = 300$ V/m. Amplitude of magnetic field will be : (a) 10^{-8} T (b) 10^{-6} T (c) 10^{-10} T (d) 10^{-7} T	1
7.	The light sources used in Young's double slit experiment are: (a) Monochromatic (b) Coherent (c) White light (d) Incoherent.	1
8.	A converging lens is used to form an image on a screen. When the upper half of the lens is covered by an opaque screen: (a) half the image will disappear. (b) incomplete image will be formed. (c) intensity of image will decrease but complete image is formed. (d) intensity of image will decrease and smaller image is formed..	1
9.	A point object is placed at centre of a glass sphere of radius 6 cm and refractive index 1.5. The distance of virtual image from surface of sphere is : (a) 2 cm (b) 4 cm (c) 6 cm (d) 12 cm	1
10.	When a ray of light enters a glass slab from air: a) its wavelength decreases b) neither wavelength nor frequency changes c) its wavelength increases d) its frequency increases	1
11.	The work function of a metal is 4.14 eV. The threshold wavelength for this metal surface is: (a) $4125 \text{ \AA}$ (b) $2062.5 \text{ \AA}$ (c) $3000 \text{ \AA}$ (d) $6000 \text{ \AA}$	1
12.	When a forward bias is applied to a p-n junction, it: (a) raises the potential barrier. (b) Reduces the majority carrier current to zero. (c) lowers the potential barrier (d) None of the above.	1

	For Questions 13 to16, two statements are given—one labeled Assertion (A) and other labeled Reason(R). Select the correct answer to these questions from the options a given below. a) If both Assertion and Reason are true and Reason is correct explanation of Assertion. b) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion. c) If Assertion is true but Reason is false. d) If both Assertion and Reason are false.	
13.	A: The mutual induction between the two coils infinitely apart is zero. R: If the mutual induction between the two coils is zero, it means that their self- inductances are also zero.	1
14.	Assertion(A): The trajectory traced by an incident particle depends on the Impact Parameter of collision. Reason(R): The Impact Parameter is the perpendicular distance of the initial velocity vector of the incident particle from the centre of the target nucleus.	1
15.	Assertion (A) – Nuclei having mass number about 60 are least stable. Reason(R) – When two or more light nuclei are combined then binding energy per nucleon will decrease.	1
16.	Assertion (A) – Photo electric current depends on the intensity of incident light. Reason(R)–Number of photo electrons emitted per second is directly proportional to the intensity of incident radiation.	1
Section-B		
17.	An electric dipole having a dipole moment of $4 \times 10^{-9} \text{ C m}$ is placed in a uniform electric field such that the dipole is in stable equilibrium. If the magnitude of the electric field is $3 \times 10^3 \text{ N/C}$, what is the work done in rotating the dipole from a position of stable equilibrium to a position of unstable equilibrium?	2
18.	Draw a sketch of an electromagnetic wave propagating along the z-direction. Write the expressions for electric and magnetic fields varying sinusoidally along X and Y axes respectively. OR Name the constituent radiation of electromagnetic spectrum which (a) Is used in satellite communication. (b) Is used for studying crystal structure. (c) Is similar to the radiations emitted during decay of radioactive nuclei? (d) Has a wavelength range between 390 nm and 770 nm.	2
19.	A solenoid with 500 turns and a cross-sectional area of 0.01 m^2 is placed in a changing magnetic field. The magnetic flux through the solenoid increases from 0.02 Wb to 0.06 Wb in 0.1 s. What is the average induced EMF in the solenoid?	2
20.	Define threshold frequency. Two metals A and B have work functions 4 eV and 10 eV, respectively. Which metal has higher threshold wavelength justify your answer?	2
21.	Draw the graph between: (i) Plate voltage and photocurrent when intensity of radiation is varying for constant	2

	frequency. (ii) Stopping potential and frequency of radiation.	
Section C		
22.	State Kirchhoff's rules. Find the value of R in the Wheatstone's bridge shown in figure if there is no deflection in the galvanometer. 	3
23.	Define drift speed of electron. Prove that the current density of a metallic conductor is directly proportional to the drift speed of electrons.	3
24.	A series LCR circuit is connected to an a.c. source having voltage $V = V_0 \sin \omega t$. Using phasor diagram, derive expressions for impedance, instantaneous current and its phase relationship to the applied voltage. Also draw graphs of V and I versus ωt for the circuit.	3
25.	Draw a neat ray diagram of refraction of light through equilateral prism and find the expression for refractive index of material of prism. Draw the graph showing the variation of angle of deviation with angle of incidence.	3
26.	A compound microscope consists of an objective lens of focal length 2.0 cm and an eyepiece of focal length 6.25 cm separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at the least distance of distinct vision (25 cm) Calculate the magnifying power of the microscope? (OR) (i) A concave lens made of material of refractive index 'n_2' is held in a reference medium of refractive index 'n_1'. Trace the path of parallel beam of light passing through the lens when: (a) $n_1 = n_2$ (b) $n_1 < n_2$ (c) $n_1 > n_2$. (ii) Find the focal length of a combination of a convex lens of focal length 30 cm and a concave lens of focal length 20 cm in contact?	3
27.	Draw the circuit diagram of full wave rectifier, state its principle and draw the input and output waveforms.	3
28.	(i) Define potential barrier in PN-Junction diode. (ii) In the following diagram, which bulb out of B_1 and B_2 will glow and which will not glow? Justify your answer in each case.	3



Section-D

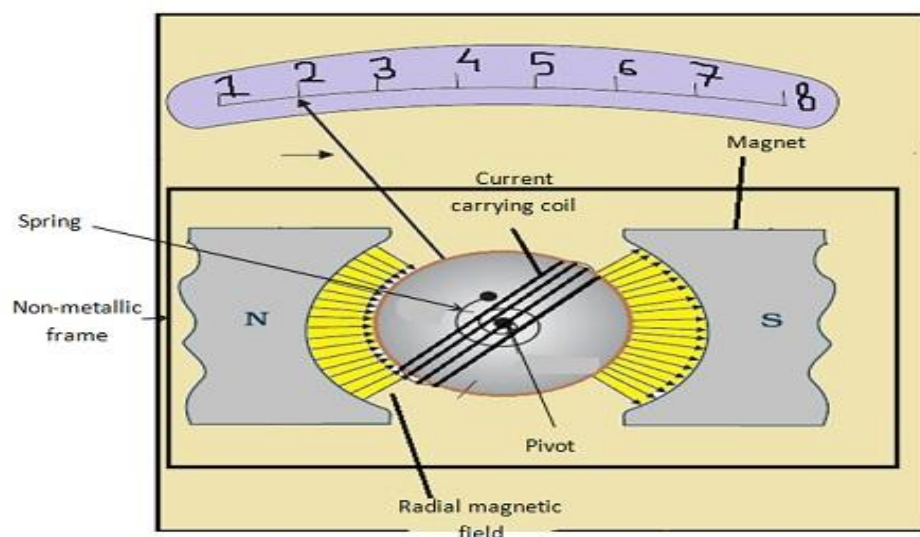
Case Study Based Questions

29 MOVING COIL GALVANOMETER

4

Moving coil galvanometer is an electromagnetic device that can measure small values of current. It is also known as Weston galvanometer. It works on the principle that when a current loop is placed in an external magnetic field, it experiences torque, and the value of torque can be changed by changing the current in the loop.

A moving coil galvanometer consists of permanent horse-shoe magnets, coil, soft iron core, pivoted spring, non-metallic frame, scale and pointer. We know that a current loop having N number of turns, and the cross sectional area A , carrying current i , when placed in and along the direction of external magnetic field B , experiences a torque given by $NiAB$.



- (i) In a moving coil galvanometer the deflection (Φ) on the scale by a pointer attached to the spring is:

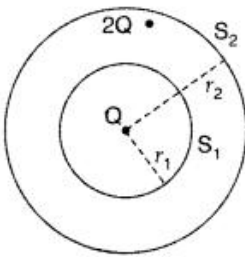
(a) $\left(\frac{NA}{kB} \right) I$

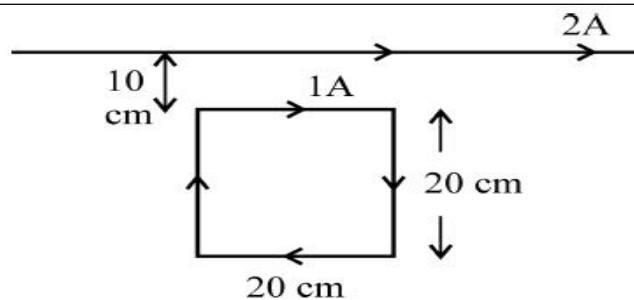
(b) $\left(\frac{N}{kAB} \right) I$

(c) $\left(\frac{NAB}{k} \right) I$

(d) $\left(\frac{NAB}{kI} \right) I$

- (ii) A moving coil galvanometer can be converted into an ammeter by:
- Introducing a resistance of large value in series.
 - Introducing a shunt resistance of small value in parallel.
 - Introducing a resistance of small value in series.
 - Introducing a resistance of large value in parallel.
- (iii) A galvanometer shows full deflection current for 5 mA. If the resistance of galvanometer is 50Ω then the resistance required to convert it into voltmeter of range 5 V is:

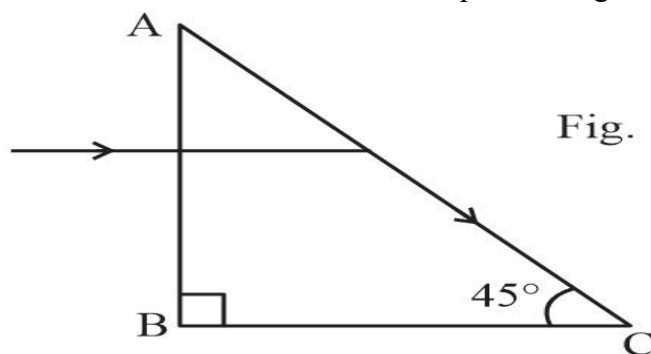
	(a) Lyman	(b) Balmer	(c) Paschen	(d) Brackett	
Section-E					
31	(i) Using Gauss Law, derive an expression for the Electric Field Intensity at any point outside a uniformly charged spherical shell of radius R and carrying charge q, with the help of suitable diagram. (ii) A uniformly charged spherical shell of 2.5 m radius has a surface charge density of $100\mu\text{C}/\text{m}^2$. Calculate the Charge on the sphere and Total electric flux passing through the sphere. OR (i) An electric dipole of dipole moment p consists of point charges $+q$ and $-q$ separated by a distance $2a$ apart. Deduce the expression for the electric field E due to the dipole at a distance x from the centre of the dipole on its axial line in terms of the dipole moment p. What is the electric field at midpoint of dipole? (ii) A sphere S_1 of radius r_1 encloses a net charge Q. If there is another concentric sphere S_2 of radius r_2 ($r_2 > r_1$) enclosing charge $2Q$, find the ratio of the electric flux through S_1 and S_2. How will the electric flux through sphere S_1 change if a medium of dielectric constant K is introduced in the space in between S_1 and S_2 in place of air? 				5
32	(i) State Ampere's circuital law and using it find magnetic field due to straight infinite current carrying wire. (ii) Draw a graph between magnetic field and perpendicular distance of observation point from the wire (iii) A long straight wire in the horizontal plane carries a current of 15 A in north to south direction. Find the magnitude and direction of magnetic field at a point 2.5 m east of the wire. OR (i) Derive the expression of force between two infinitely long parallel current carrying conductors, hence define one ampere of current. (ii) A square loop of side 20 cm carrying current of 1 A is kept near an infinite long straight wire carrying a current of 2 A in the same plane as shown in the figure. Calculate the magnitude and direction of the net force exerted on the loop due to the current carrying conductor.				5



33

- (i) Using Huygens wave theory of light, derive Snell's law of refraction. What is the shape of the wave front in each of the following cases:
- Light emerging out of a convex lens when a point source is placed at its focus.
 - The portion of the wave front of light from a distant star intercepted by the Earth
- (ii) A light ray entering a right-angled prism undergoes refraction at the face AC as shown in Fig.

What is the refractive index of the material of the prism in fig.



OR

- Draw a labelled diagram of image formation by a compound microscope and write the expression for magnification produced by compound microscope
- In Young's double-slit experiment using monochromatic light of wavelength λ , the intensity of light at a point on the screen where path difference is λ , is K units. What is the intensity of light at a point where path difference is $\lambda/3$?

5